

Section 3.8 Exercise Answers

1. $f^{-1}(x) = x - 3$.
2. $f^{-1}(x) = x - 5$.
3. $f^{-1}(x) = 2 - x$.
4. $f^{-1}(x) = 3 - x$.
5. $f^{-1}(x) = \frac{x - 7}{11}$.
6. $f^{-1}(x) = \frac{3 - 4x}{5x - 2}$.
7. $[-7, \infty)$; $f^{-1}(x) = \sqrt{x} - 7$
8. $[6, \infty)$; $f^{-1}(x) = \sqrt{x} + 6$.
9. $[0, \infty)$; $f^{-1}(x) = \sqrt{x + 5}$.
10. Ⓐ $f(g(x)) = x$; $g(f(x)) = x$ Ⓑ That f and g are inverses.
11. $f(g(x)) = x$; $g(f(x)) = x$. So f and g are inverses.
12. $f(g(x)) = x$; $g(f(x)) = x$. So f and g are inverses.
13. Not one-to-one
14. Not one-to-one
15. $f(0) = 3$
16. $y = 0$ when $x = 2$
17. $f^{-1}(0) = 2$
18. $f^{-1}(x) = 0$ when $x = 3$
19. $f^{-1}(x) = \sqrt{x} + 4$
20. $f^{-1}(x) = \sqrt{x} - 2$
21. $f^{-1}(x) = \sqrt{x + 3} - 1$
22. $f^{-1}(x) = (2 - x)^2 - 3$
23. $f^{-1}(x) = -\sqrt{\frac{x - 5}{3}}$
24. $f^{-1}(x) = \sqrt{12 - x}$
25. $f^{-1}(x) = \sqrt{9 - x}$
26. $f^{-1}(x) = \sqrt{\frac{x - 4}{2}}$
27. $f^{-1}(x) = \sqrt[3]{x - 5}$
28. $f^{-1}(x) = \sqrt[3]{\frac{x - 1}{3}}$

$$29. f^{-1}(x) = \sqrt[3]{4-x}$$

$$30. f^{-1}(x) = \sqrt[3]{\frac{4-x}{2}}$$

$$31. f^{-1}(x) = \frac{x^2-1}{2}$$

$$32. f^{-1}(x) = \frac{3-x^2}{4}$$

$$33. f^{-1}(x) = \frac{(x-9)^2+4}{4}$$

$$34. f^{-1}(x) = \frac{(x-5)^2+8}{6}$$

$$35. f^{-1}(x) = \left(\frac{x-9}{2}\right)^3$$

$$36. f^{-1}(x) = (3-x)^3$$

$$37. f^{-1}(x) = \frac{2-8x}{x}$$

$$38. f^{-1}(x) = \frac{4x+3}{x}$$

$$39. f^{-1}(x) = \frac{7x-3}{1-x}$$

$$40. f^{-1}(x) = \frac{7x+2}{1-x}$$

$$41. f^{-1}(x) = \frac{5x-4}{4x+3}$$

$$42. f^{-1}(x) = \frac{2x-1}{5x+5}$$